Numerical Optimization and Nonlinear Models for Heart-Disease Classification

Jacobo Romero Pablo González Roger Baiges
EPFL, School of Computer and Communication Sciences
CS-433 Machine Learning · Project 1 · Fall 2026

***Abstract*—We study how data representation, numerical optimization, and model capacity affect heart-disease classification in BRFSS survey responses. Under a NumPy-only modeling constraint, we implement six linear and logistic methods, investigate neural networks, and extend the comparison to gradient-boosted trees. Encoding categories and missingness yields a strong logistic baseline; greater neural capacity provides limited gains. Repeated cross-validation favors boosted trees, while engineered features produce the highest recorded F1 of 0.4474. We distinguish these local comparisons from the team's reported public competition F1 of 0.456.**

I. Problem and Data Representation

Our task is to classify self-reported coronary heart disease or myocardial infarction (`_MICHD`) from the 2015 Behavioral Risk Factor Surveillance System (BRFSS) [1]. The course release contains 328,135 labeled respondents, 109,379 test respondents, and 321 predictors. Only 8.83% of training labels are positive: predicting every respondent negative already gives 91.17% accuracy, but zero positive-class F1. We therefore prioritize F1 and use ROC AUC as a threshold-independent ranking measure.

Preserving questionnaire semantics: Almost 45% of entries are missing, often because a question is inapplicable or belongs to an optional module. Numeric codes are not necessarily quantities: `PHYSHLTH=88` means zero days of poor physical health, whereas 77 and 99 mean unknown and refused. Codebook-based rules remove 14 survey-design fields, two exact copies, and 14 mixed-unit fields with computed replacements. We convert units and fill implied numeric skips, leaving 56 numeric and 235 categorical predictors. Categorical unknown and refused answers retain their identities.

A representation suited to linear models: Numeric variables receive training-median imputation and missingness indicators. Categorical levels observed at least 30 times are one-hot encoded, with a separate missing category; rare or unseen levels map to all zeros. Standardization yields approximately 1,530 columns. All learned statistics are fitted within each training fold. Figure 1 illustrates the importance of encoding; it is an exploratory diagnostic, not the repeated-CV comparison below.

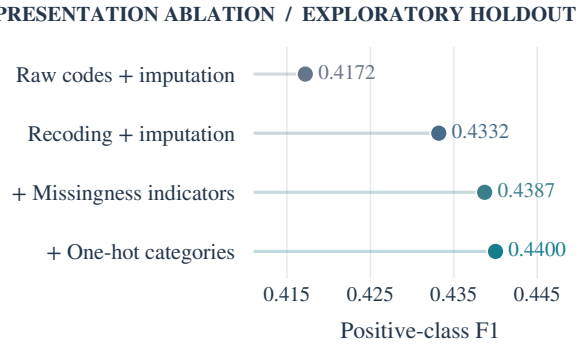


Figure 1. Exploratory logistic-regression ablation on one 80/20 split. F1 uses a validation-optimized threshold and is optimistic. Recoding also removes survey-design fields and maps categorical unknowns to missing; the final pipeline instead retains these categorical codes.

II. Numerical Methods and Model Capacity

Objectives and optimization: For a design matrix X including an intercept, labels $y \in \{0,1\}^N$, and scores $z = Xw$, we minimize

$$L_{\text{MSE}}(w) = \frac{\|Xw - y\|_2^2}{2N}, \tag{1}$$

$$L_{\log}(w) = \frac{1}{N}\sum_{i=1}^{N}\left[\log(1+e^{z_i}) - y_i z_i\right]. \tag{2}$$

Ridge and regularized logistic regression add $\lambda\|w\|_2^2$; gradient descent (GD) updates $w \leftarrow w - \gamma\nabla L(w)$. We compare MSE GD, single-sample stochastic GD (SGD), least squares, ridge, logistic GD, and regularized logistic GD, all implemented in NumPy.

Conditioning and numerical stability: One-hot dependencies make $X^\top X$ singular; least squares uses a pseudoinverse and ridge solves $(X^\top X + 2N\lambda I)w = X^\top y$. The measured $\lambda_{\max}(X^\top X/N) \simeq 63.3$ gives MSE GD's stability condition $\gamma < 2/63.3$. We use $\gamma = 0.02$ for 100 steps, and $\gamma = 0.1$ for 300 logistic steps. SGD takes 10^6 single-sample updates at $\gamma = 5 \cdot 10^{-6}$; rare standardized categories produce large row norms. Both regularized presets use $\lambda = 10^{-3}$. Stable `logaddexp` evaluates logistic loss. MSE solvers share a hypothesis class; finite budgets affect their numerical solutions.

Table I
FIVE-FOLD COMPARISON OF NUMERICAL METHODS. ALL ROWS EXCEPT "RAW INPUTS" SHARE THE ENCODED DESIGN. SPREAD IS FOLD STANDARD DEVIATION, NOT A CONFIDENCE INTERVAL.

Method	F1 (mean ± SD)	AUC
Raw inputs / LS	0.4095 ± 0.0019	0.845
MSE / GD	0.4289 ± 0.0022	0.858
MSE / SGD	0.4255 ± 0.0033	0.855
Least squares	0.4301 ± 0.0018	0.857
Ridge	0.4297 ± 0.0031	0.858
Logistic / GD	**0.4390** ± 0.0027	0.864
L2 logistic / GD	0.4388 ± 0.0022	0.864

Neural networks as a capacity test: Exploratory multilayer perceptrons (MLPs) use one hidden ReLU layer, Adam optimization [2], and validation-based early stopping. Widths 64 and 256 achieve holdout F1 of 0.4341 and 0.4402; the latter uses dropout 0.3 [3]. Reported overfitting after one to two epochs motivates regularization rather than unbounded expansion. A later MLP achieves repeated-CV F1 of 0.4399. These experiments test whether learned nonlinear interactions improve on a well-encoded additive classifier; the observed gains are limited.

A nonlinear extension with explicit interactions: Motivated by gradient boosting [4], [5], we implement histogram trees in NumPy, with learned missing-value routing. A tuned ensemble uses depth 5, 550 trees, learning rate 0.05, 64 bins, leaf penalty 10, and row/feature fractions 0.7/0.3, averaging logits from model seeds 0–2. A later variant uses 2,000 trees at rate 0.02 and adds eight response-count features plus categorical target-rate encoding, smoothed with 50 pseudo-rows at the training prevalence. Target statistics are learned only on training folds. The five-member feature ensemble combines three seeds with two additional variants: feature fraction 0.2 and minimum child Hessian 1.

III. Evaluation and Results

Protocol and decision threshold: Table I compares the six required methods on the same five shuffled folds (seed 42), with thresholds optimized on each model's training scores. The advanced comparison uses split seeds 42–44, five folds each; each scored fold takes its threshold from the other folds' out-of-fold predictions. Thresholds around 0.20–0.24 outperform the default 0.5 in this imbalanced task. Hyperparameters and blend weights are selected using these validation data, so this is not fully nested model-selection CV.

Representation and loss matter before complexity: For least squares, encoding increases F1 from 0.4095 to 0.4301. Logistic GD reaches 0.4390 and improves on least squares on every fold; the tested L2 penalty adds no benefit. The exploratory encoding sequence in Fig. 1 supports preserving missingness and categories. However, once one-hot encoding is present, numeric cleaning alone changes paired five-fold F1 by only −0.0003; its semantic correctness should not be confused with a measurable incremental gain.

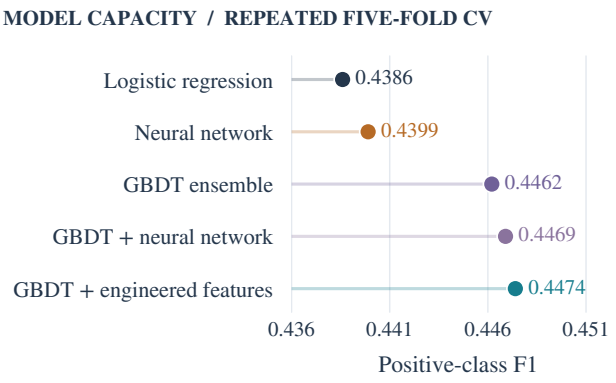


Figure 2. Recorded means over 15 validation folds, with thresholds selected from other folds' OOF scores. Points are selected configurations; no error bars are inferred from unavailable fold records. The later MLP is a separate experiment from the exploratory networks in Section II.

Nonlinearity helps; the final increments are uncertain: In the repeated comparison, logistic regression reaches 0.4386, the MLP 0.4399, and the tuned tree ensemble 0.4462 (AUC 0.8703). The recorded Nadeau–Bengio corrected paired comparison [6] estimates a tree–logistic F1 gain of 0.0077, with a 95% interval $[0.0038, 0.0115]$. Adding the MLP to the trees reaches 0.4469; engineered features and ensembling reach 0.4474 (Fig. 2). The latter gain is only about 0.0011 over the tuned ensemble (corrected $t = 1.51$). These selected-model comparisons support the larger tree gain more strongly than the final refinements; they do not account for the full model search. The team's best reported public AIcrowd F1 is 0.456, kept separate from the local CV results.

IV. Discussion and Reproducibility

Our strongest finding is the progression from meaningful representation to a well-optimized logistic baseline, then to a nonlinear model with a larger measured gain. Extra neural capacity and ensemble diversity do not automatically improve generalization. Importantly, all 773 training respondents answering `HAREHAB1` are positive: this post-infarction rehabilitation question [1] leaks diagnosis information. Retaining it aids the benchmark but precludes interpreting the scores as prospective clinical prediction.

Executable numerical methods, transforms, and GBDT recipes accompany this report. Results come from archived summaries; MLP source and fold artifacts are absent from this checkout. The current `run.py` refits the earlier three-member ensemble. Linking the 0.456 submission to its exact configuration and verifying identical predictions remain necessary for final reproducibility.

References

[1] Centers for Disease Control and Prevention, "Behavioral Risk Factor Surveillance System: 2015 Codebook Report," U.S. Department of Health and Human Services, Tech. Rep., 2016. [Online]. Available: https://www.cdc.gov/brfss/annual_data/2015/pdf/codebook15_llcp.pdf

[2] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *International Conference on Learning Representations*, 2015. [Online]. Available: https://arxiv.org/abs/1412.6980

[3] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *Journal of Machine Learning Research*, vol. 15, pp. 1929–1958, 2014. [Online]. Available: https://jmlr.org/papers/v15/srivastava14a.html

[4] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794. [Online]. Available: https://arxiv.org/abs/1603.02754

[5] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems*, vol. 30, 2017. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html

[6] C. Nadeau and Y. Bengio, "Inference for the generalization error," *Machine Learning*, vol. 52, no. 3, pp. 239–281, 2003. [Online]. Available: https://link.springer.com/article/10.1023/A:1024068626366